When Better QAOA Schedules Depend on the Simulator: Exact and Cross-Backend Rank Reversals on QOBLIB

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Abstract

Approximate tensor-network simulators extend variational quantum algorithms beyond the statevector limit, but rankings are often reported at one accuracy setting. We test ranking stability for depth-15 QAOA schedules on real QOBLIB Maximum Independent Set benchmarks. On a 186-vertex graph reduced to 55 qubits, the released Aer MPS setting gives 101 best-known-solution (BKS) hits for a transferred nonlinear ramp and 41 for the published linear ramp in 15,000 shots each; tightening only the cutoff reverses the BKS ranking. We then freeze a five-case exact replication with five MPS settings, three schedules, two qubit orderings, and Aer and cuTensorNet: 300 dense backend rows, including 240 new executions. For schedules i, j, the event-effect error obeys $|\widetilde{\Delta} - \Delta| \leq \mathrm{TV}_i + \mathrm{TV}_j$. All 100 tested backend/setting/ordering inequalities hold. The certificate $(\mathrm{TV}_i + \mathrm{TV}_j)/|\Delta| < 1$ covers 77 cohorts and all 77 preserve the exact sign; outside it, only 14 of 23 do (Fisher exact $p = 4.30 \times 10^{-7}$). Overall sign correctness is 91/100 and cross-backend sign agreement is 45/50, with all nine sign failures concentrated in the 24-qubit, smallest-margin case. Thus rare-event rankings require observable-level error budgets rather than a single nominal MPS setting.

1 Introduction

QAOA alternates problem and mixer Hamiltonians with variational angles chosen by a classical optimization loop [1]. Parameter transfer amortizes that loop by learning angles on donor instances and deploying them on larger acceptors [2, 3]. This strategy is especially attractive when the target circuit is too large for exact statevector simulation. In practice, such targets are often evaluated with approximate tensor networks, whose runtime and accuracy are controlled by bond dimensions and truncation thresholds [10, 12].

The optimization literature usually asks which QAOA schedule is better. The simulation literature asks how runtime and aggregate error scale with an approximation parameter. We ask a question at their intersection: *does the simulator preserve the ordering of QAOA schedules on the observable used to declare a winner?* This is not guaranteed. The probability of an optimum may live in a small tail of the output distribution; a small state approximation error can therefore cause a large relative error in that rare-event probability without visibly changing average energy or feasibility.

We study this issue on the Quantum Optimization Benchmarking Library (QOBLIB), a curated, submission-oriented framework designed for reproducible comparison of quantum and classical optimization methods [8]. We reproduce the released depth-15 Aer/MPS pipeline for the `es60fst02` Maximum Independent Set (MIS) instance, including graph reduction, Hamiltonian construction, sample unfolding, and a validity-only filter. We compare its published linear-ramp schedule with a low-dimensional nonlinear ramp selected on other full ES60FST instances.

Our contributions are:

1. a train-validation-test protocol over four complete QOBLIB graphs, ending in a 55-qubit, depth-15 blind benchmark with no repair or target retuning;
2. an equal-budget comparison of evolutionary and random schedule search, showing that the random-search champion, not the evolutionary operator, transfers best;
3. 15 paired MPS jobs at the published simulator setting, with bootstrap intervals and an exact paired randomization test;
4. a bond/threshold sensitivity audit that identifies a truncation-induced BKS rank reversal while near-optimal and feasible-mass rankings remain stable; and
5. dense exact adjudication on five real QOBLIB kernels (3–24 qubits), followed by a frozen 300-row Aer/cuTensorNet replication; and
6. an observable-level TVD certificate that preserves the exact ranking in all 77 certified cohorts and isolates all nine observed failures outside the certified region.

We do not claim quantum advantage, an exact simulation of the 55-qubit state, or universal superiority of the nonlinear ramp. The main claim is about benchmark validity: schedule rankings based on rare optimum hits can depend on the truncation setting, backend implementation, and qubit ordering even when all circuits are exactly equivalent before truncation.

2 Background and related work

Parameter transfer and schedule reduction. QAOA parameters concentrate across related random graphs, mo-

Table 1: Full QOBLIB instances and quantum kernels at maximum retained degree four. V/E are original vertices/edges; n_q/m_q are reduced vertices/edges.

Instance	V	E	n_q	m_q	Role / BKS
es60fst01	123	159	15	21	train / 60
es60fst03	113	142	12	16	train / 55
es60fst04	162	238	45	69	validation / 78
es60fst02	186	280	55	91	blind test / 88

tivating donor selection, representation learning, and direct parameter transfer [2, 3]. Weighted-problem studies show that rescaling phase angles can restore transfer when coefficient distributions change [4]. Linear-ramp QAOA compresses $2p$ angles to two increments and has been studied at larger depths [5]. Recent constrained-QAOA work promotes penalty schedules and piecewise ramps to variational parameters [6]. Our nonlinear ramp is intentionally modest; it is a vehicle for testing ranking stability, not the central theoretical novelty.

Evolutionary variational optimization. Evolutionary methods are natural for nonconvex, noisy quantum objectives [7]. We use a fixed-population genetic search and compare it with uniform random search under exactly the same number of candidate evaluations. The negative result—matched random search finds the stronger transferring schedule—prevents attributing the observed gain to evolution itself.

Approximate tensor-network simulation. MPS simulation compresses a many-qubit state by truncating Schmidt spectra; accuracy depends on circuit ordering, entanglement growth, the maximum bond dimension, and discarded-weight policy [10]. Recent work studies tensor-network surrogates for variational circuits and the compute-accuracy frontier of approximate simulators [11, 12]. Exact tensor-network sampling methods have also reached structured QAOA circuits beyond conventional statevector sizes [13]. These studies motivate careful convergence analysis, but they do not establish that optimization algorithm rankings are invariant under the approximation controls used in a submitted benchmark.

3 Benchmark protocol

3.1 Instances and frozen split

Table 1 lists the four full QOBLIB ES60FST MIS instances. The split was frozen before evaluation of any new schedule on `es60fst02`. Best-known values are exact Gurobi optima in the QOBLIB submission records.

We use the released reduction implementation: sequential exact degree-0/1 rules, exact degree-2 folding, recorded pruning of vertices above degree four, and a final exact cleanup. Pruning is heuristic; all decisions are recorded so samples can be unfolded. On `es60fst02`, 37 vertices are forced selected, 38 forced excluded, 28 folds occur in four batches, and five vertices are heuristically pruned. The published workflow also uses this 55-qubit kernel and reaches BKS 88 under Aer MPS [9].

3.2 Hamiltonian, circuit, and schedules

For graph $G = (V, E)$, the MIS QUBO is

$$C(x) = -\sum_{i\in V} x_i + \lambda \sum_{(i,j)\in E} x_i x_j, \qquad \lambda = 1.5. \tag{1}$$

After $x_i = (1 - Z_i)/2$, QAOA alternates the Ising cost operator and $H_M = \sum_i X_i$. Both methods use depth $p = 15$, maximum-quadratic-coefficient Hamiltonian scaling from the released code, an initial $|+\rangle^{\otimes n}$ state, and identical measurement conventions. The untranspiled test circuit contains 1,365 RZZ, 825 RZ, 825 RX, 55 Hadamard, and 55 measurement operations.

The published linear ramp is

$$\beta_k = 0.7\frac{p-k+1}{p}, \qquad \gamma_k = 0.4\frac{k}{p}. \tag{2}$$

The candidate family generalizes the ramp with two powers:

$$\beta_k = \Delta_\beta \left(\frac{p-k+1}{p}\right)^{a_\beta}, \tag{3}$$

$$\gamma_k = \Delta_\gamma \left(\frac{k}{p}\right)^{a_\gamma}. \tag{4}$$

The four-dimensional genome lies in $\Delta_\beta \in [0.15, 1.20]$, $\Delta_\gamma \in [0.05, 1.00]$, and $a_\beta, a_\gamma \in [0.30, 2.50]$. The frozen random-search champion is $(0.64247, 0.75939, 1.77679, 0.99172)$; the evolutionary champion is $(0.51750, 0.77197, 1.07734, 1.75435)$.

3.3 Search, selection, and metrics

Three independent search replicates use 20 candidates for six generations, or 120 candidate evaluations per method. Each candidate receives 256 shots on each training instance. Random search receives exactly 120 independently drawn candidates. A finite-shot robust score combines Wilson lower bounds on BKS, BKS-1, and feasible probabilities with feasible quality mass; its mean and worst-instance values are combined 3:1. One validation pass on `es60fst04` selects one champion per search method. Test outcomes are never used for schedule selection.

For every reduced sample, the released decoder reverses folds and forced decisions. We retain the full sample only if it is already an independent set. There is no constraint repair, greedy fill, local search, or archived-solution fallback. We report

- BKS hit probability $\mathbb{P}(|S| \geq 88)$;
- near-BKS probability $\mathbb{P}(|S| \geq 87)$;
- feasible probability; and

- quality mass $\mathbb{E}[\mathbf{1}_{\text{feasible}} \min(|S|/88, 1)]$.

The confirmatory run uses 15 paired simulator seeds and 1,000 shots per method and seed. Paired mean differences receive a 20,000-draw bootstrap interval and an exact two-sided sign-flip test over all 2^{15} paired sign assignments.

3.4 Independent backend and classical controls

We export the frozen native circuits to NVIDIA cuTensorNet 26.6 [14]. Direct contraction reproduces Qiskit statevectors for both schedules on both 12- and 15-qubit kernels with fidelity one to displayed precision. The 55-qubit exact sampler fails during network preparation, so all completed large-circuit results are explicitly approximate MPS results. We use a Fiedler-vector qubit ordering, exact-equivalent before truncation, that lowers edge bandwidth from 47 to 10 and maximum linear cut from 42 to 9. We test bond 128 with discarded-weight cutoffs 10^{-3} and 10^{-4} using 5,000 shots per schedule. Samples are inverse-permuted before the unchanged decoder is applied.

Classical controls operate on the original full graph and on the released 55-variable kernel. An exact binary MILP is solved with SciPy/HiGHS at zero MIP gap. A randomized minimum-residual-degree heuristic repeatedly selects a minimum-degree vertex with random tie breaking, deletes it and its neighbors, and uses no archived solution. Both heuristic variants receive 15,000 starts.

3.5 Simulator settings and audit

The primary setting exactly follows the submission: Qiskit Aer MPS, maximum bond dimension 64, truncation threshold 10^{-3}, and native RZ/RZZ/RX circuits with no hardware-basis transpilation. This detail matters because a gate decomposition changes when MPS truncation is applied. A first transpiled pass was preserved as a protocol audit and excluded from the main result.

We then hold circuits and shots fixed while testing thresholds 10^{-3}, 3×10^{-4}, and 10^{-4} at bond 64, plus bonds 64 and 96 at thresholds 10^{-3} and 10^{-4}. Audit points use 10,000 shots per principal method; the primary $10^{-3}/64$ point aggregates 15,000 shots.

To calibrate the direction and magnitude of approximation error, we also run complete statevectors on the real 12- and 15-qubit training kernels. For both principal schedules we compare exact probabilities with MPS states at bond dimensions 16, 32, 64, and 128 and thresholds from 10^{-3} to 10^{-6}. We report state fidelity, total-variation distance, Jensen-Shannon divergence, and signed errors in BKS, near-BKS, feasibility, and quality mass. These are probability-exact comparisons with no measurement noise.

3.6 Exact external adjudication and cross-backend ladder

After the 55-qubit study, we froze a separate exact cross-case experiment on five real QOBLIB instances: `karate`, `chesapeake`, `football`, `ibm32`, and `aves-sparrow-social`. Their frozen reductions contain 3, 7, 7, 18, and 24 qubits. We evaluate the published LR, prior evolutionary, and prior matched-random depth-15 schedules under sorted and spectral qubit orderings. Dense statevectors enumerate every outcome; the QOBLIB unfolding and feasibility rules are compiled into bit literals and accumulated without shots or repair.

The independent ladder fixes five points before execution: bond/cutoff $(64, 10^{-3})$, $(128, 10^{-4})$, $(128, 10^{-12})$, $(1024, 10^{-4})$, and $(1024, 10^{-5})$. The six native circuits are exported as hashed QPY files to cuTensorNet 26.6. Both backends return dense approximate states, which are normalized and compared with the six exact states by fidelity, total variation, and application metrics. The design contains exactly 5 cases $\times$ 5 settings $\times$ 3 schedules $\times$ 2 orderings $\times$ 2 backends = 300 dense rows. The completed 60-row 24-qubit cohort was reused only after SHA-256 validation; the other 240 rows were executed after the cross-case protocol was frozen. The primary estimand is the sign of matched-random-minus-LR BKS probability; incomplete cohorts are not interpreted.

For BKS event A, exact distributions p_i, p_j, and approximate distributions q_i, q_j, we pre-specify the observable-rank certificate

$$|[q_i(A) - q_j(A)] - [p_i(A) - p_j(A)]| \leq \mathrm{TV}(q_i, p_i) + \mathrm{TV}(q_j, p_j). \tag{5}$$

The exact sign is certified whenever $|p_i(A) - p_j(A)|$ exceeds the right-hand side. Its derivation and complete cohort table are provided in the Supplement.

4 Results

4.1 Search method is not the source of the gain

On the 45-qubit validation kernel, the evolutionary champion raises feasible probability from 0.561 to 0.930 but lowers BKS probability from 0.040 to 0.017. The matched-random champion attains 0.941 feasibility, 0.051 BKS, and 0.475 near-BKS. This result was fixed before the corrected blind pass. It shows that the nonlinear family contains transferable schedules, but the evolutionary operator does not outperform a matched random search at this budget.

4.2 Published MPS setting

Table 2 and Fig. 1 summarize 45 method-jobs on the blind instance. All methods find BKS 88 in every aggregate. The random-search nonlinear ramp improves all four reported rates relative to the published linear ramp. It produces 101 BKS hits versus 41, and its BKS advantage is positive in 13 seeds with two ties and no losses. The estimated shot count for a 95% chance of at least one BKS sample, $\lceil \log(0.05)/\log(1 - p_{\text{BKS}}) \rceil$, falls from 1,095 to 444.

For matched random minus published LR, the paired BKS difference is 0.00400 (bootstrap 95% CI

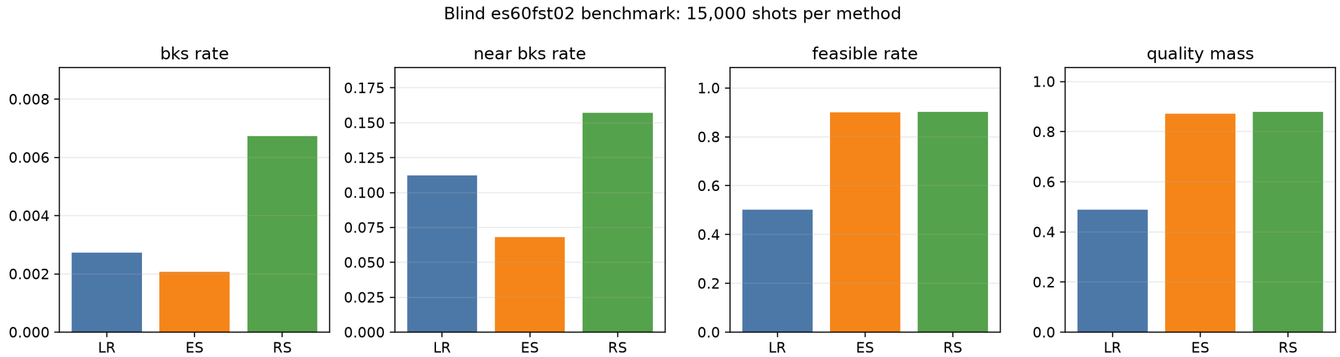

Figure 1: Blind metrics at the released Aer/MPS setting. LR is the published linear ramp; ES and RS are the frozen evolutionary- and random-search nonlinear ramps. Bars aggregate 15 paired jobs and 15,000 shots per method.

Table 2: Blind `es60fst02` results at bond 64 and threshold 10^{-3}; 15,000 shots per method.

Method	BKS	BKS-1	Feasible	Quality
Published LR	.00273	.11213	.50127	.48962
Evolutionary	.00207	.06813	.89987	.87202
Matched random	**.00673**	**.15700**	**.90180**	**.87865**

$[0.00247, 0.00553]$, sign-flip $p = 0.000244$). Corresponding differences are 0.04487 for near-BKS, 0.40053 for feasibility, and 0.38904 for quality mass; all three have sign-flip $p = 6.10 \times 10^{-5}$. The evolutionary champion improves feasibility and quality mass but not BKS or near-BKS, reinforcing the negative optimizer result.

4.3 Truncation-induced rank reversal

Figure 2 shows the central finding. At bond 64 and threshold 10^{-3}, the nonlinear ramp has the larger BKS rate. At 3×10^{-4} it still leads, 0.0161 to 0.0116. At 10^{-4} the order reverses: published LR reaches 0.0153 and the nonlinear ramp 0.0109. The shot-noise standard error of a rate near 0.01 with 10,000 shots is approximately 0.001, smaller than the observed 0.0044 gap.

The same threshold tightening does not reverse the broader metrics. At $10^{-4}/64$, nonlinear versus LR is 0.1621 versus 0.1546 for near-BKS and 0.7351 versus 0.4809 for feasibility. At $10^{-4}/96$ it is 0.1808 versus 0.1710 near-BKS and 0.7121 versus 0.4799 feasible. The optimum-hit tail is thus more ranking-sensitive than near-optimal or feasible mass.

Increasing the bond cap alone does not explain the effect. At threshold 10^{-3} and bond 96, nonlinear still leads BKS 0.0070 to 0.0019, near-BKS 0.1661 to 0.1188, and feasibility 0.8925 to 0.5055. At threshold 10^{-4} and bond 96, the BKS comparison is 0.0134 versus 0.0152, again favoring LR, while near-BKS and feasibility continue to favor nonlinear. The threshold is the dominant control in the tested range.

Runtime makes this audit consequential rather than cosmetic. At bond 96, tightening the threshold from 10^{-3} to 10^{-4} increases the two-circuit batch from about 38 seconds to about 347 seconds on the local CPU. An attempted bond-128, 10^{-6} three-method batch exceeded 20 minutes and was terminated without producing a result. Single-setting reports therefore create a real temptation to use a cheap but unconverged point.

4.4 Exact-state calibration isolates threshold bias

The small-kernel calibration supplies ground truth for the same schedules and reduction family. On `es60fst01`, exact BKS probabilities are 0.68610 for LR and 0.92898 for the nonlinear ramp. At threshold 10^{-3}, MPS shifts them by $+0.01577$ and $+0.03417$, respectively. On `es60fst03`, exact BKS probabilities are 0.73583 and 0.95530; corresponding MPS biases are $+0.00744$ and $+0.01849$. Thus, the released threshold overstates BKS probability for both schedules but more strongly for the nonlinear ramp.

Figure 3 shows convergence. At 10^{-3}, total-variation distance is 0.0518/0.0509 for LR and 0.0742/0.0702 for nonlinear on the two kernels. At 10^{-6} all four state fidelities exceed 0.99983, total variation is below 0.00266, and absolute BKS error is below 0.00023. For shared thresholds, results are numerically identical at bonds 32, 64, and 128. Bond 16 differs only for the nonlinear `es60fst01` circuit at 10^{-4}, where total variation changes by less than 0.0005. Thus, once the cap reaches 32, the discarded-weight threshold controls the error. The nonmonotonic signed BKS error for LR also shows why convergence of a rare observable cannot be inferred solely from a monotonically improving global state metric.

4.5 Cross-case replication and rank certificate

The five exact matched-random-minus-LR BKS effects are $+0.086412$ for `karate`, -0.134214 for `chesapeake`, $+0.019269$ for `football`, -0.246123 for `ibm32`, and -0.012139 for `aves-sparrow-social`. The design therefore spans both effect signs, 3–24 qubits, and a 20-fold

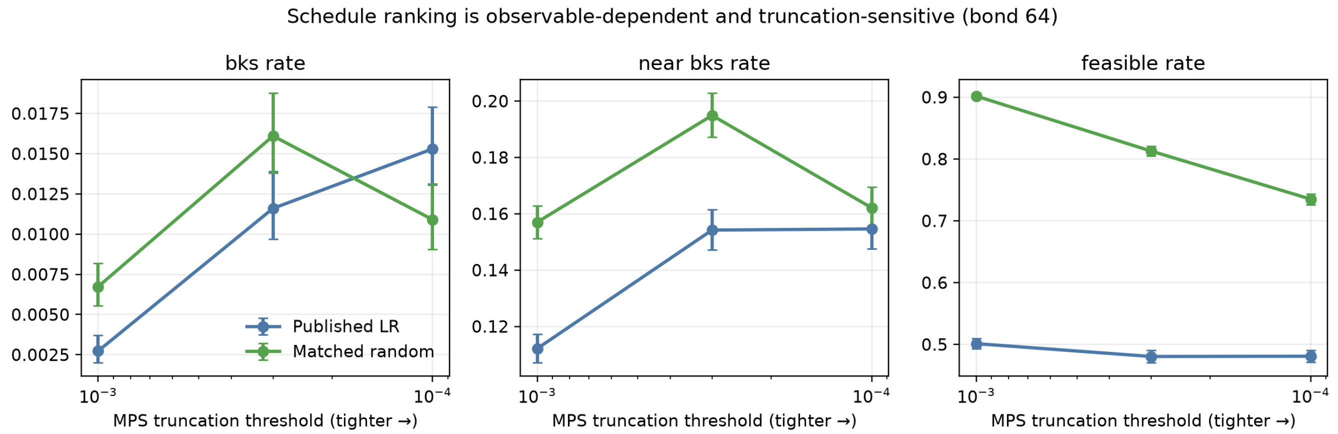

Figure 2: Observable-dependent convergence at fixed bond dimension 64. Tightening the threshold reverses the BKS ranking between 3×10^{-4} and 10^{-4}, while near-BKS and feasible rankings remain unchanged. The 10^{-3} point uses 15,000 shots; the other points use 10,000 shots per method.

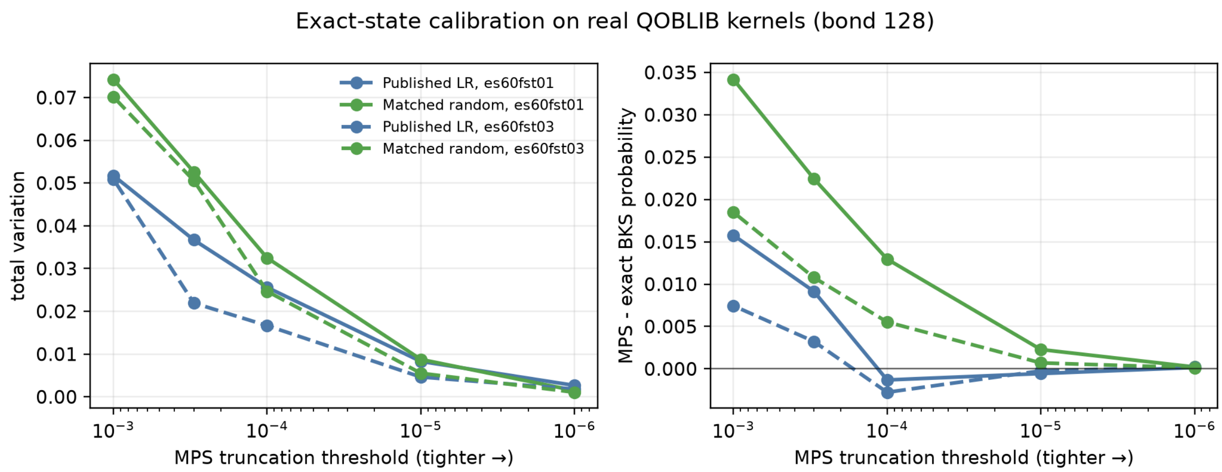

Figure 3: Exact-state MPS calibration on the 15-qubit `es60fst01` and 12-qubit `es60fst03` kernels at bond 128. Left: total-variation distance from the exact output distribution. Right: signed error in BKS probability. Tightening the truncation threshold drives both errors toward zero, but the rare-event bias is schedule-dependent and need not be monotone.

range of nonzero effect magnitudes.

Figure 4 summarizes all 100 backend/setting/ordering cohorts. Equation 5 holds in every cohort; the largest numerical excess of observed error over its bound is 6.6×10^{-16}, consistent with floating-point accumulation. Overall, 91/100 approximate matched effects have the exact sign and Aer/cuTensorNet agree in 45/50 matched cohorts. The exact margin exceeds the TVD error budget in 77/100 cohorts. Every one of these 77 certified cohorts is sign-correct (Clopper–Pearson 95% lower bound 0.953), whereas 14/23 uncertified cohorts are correct. This contrast is descriptive but large (two-sided Fisher exact $p = 4.30 \times 10^{-7}$); the guarantee itself comes from Eq. 5, not from the test.

The per-case pattern supports an effect-margin explanation rather than a qubit count rule. `karate` and `chesapeake` are universally certified from the released point; `football` first becomes universal at the confirm point and `ibm32` at bond 1024/cutoff 10^{-4}. All 80 cohorts on these four cases preserve the exact sign, while 72/80 are certified, leaving eight conservative false negatives of the certificate. The smallest-margin 24-qubit case accounts for all nine sign failures and all five cross-backend sign disagreements. Its tightest setting is sign-correct in all four backend/order combinations but is still not universally TVD-certified, illustrating that a sufficient certificate need not be necessary.

4.6 Exact 24-qubit cross-backend adjudication

The exact 24-qubit BKS probabilities are 0.323277 for LR, 0.311138 for the matched-random schedule, and 0.425256 for the evolutionary schedule. They are invariant to the two

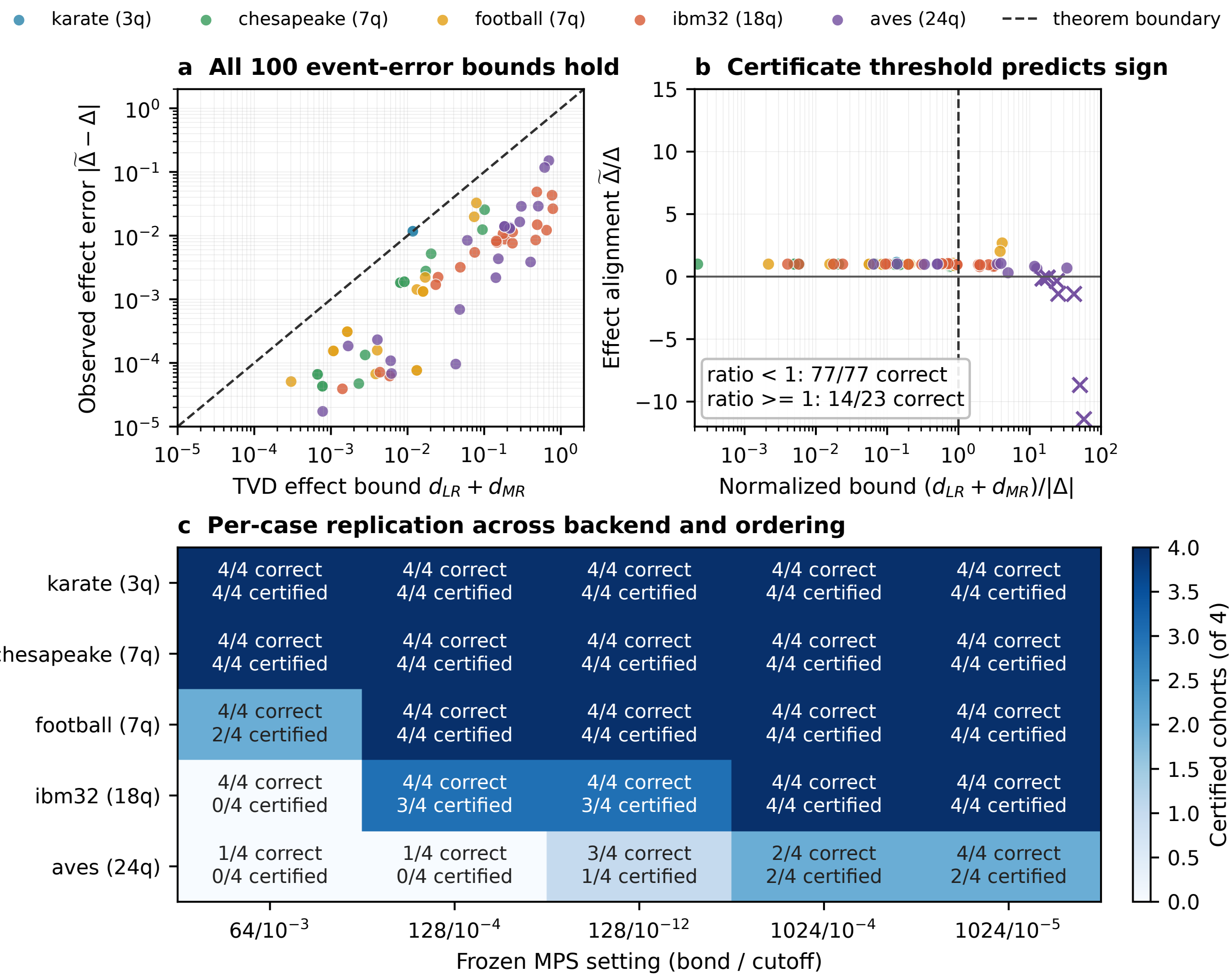

Figure 4: Frozen five-case exact replication. (a) Observed matched-effect error never exceeds the sum of schedule TVDs. (b) The normalized TVD budget provides a deterministic sign certificate below one; crosses mark wrong-sign cohorts. (c) Correct and certified cohorts among the four backend/order combinations at each case and MPS setting.

exact-equivalent qubit orderings. Thus the exact matched-random-minus-LR effect is -0.012139: the small positive advantages seen at several approximate settings have the wrong sign.

Figure 5 gives all ten setting/ordering locations, comprising 20 backend cohorts. At the released point Aer reports $+0.138479$ and $+0.105273$ for sorted and spectral ordering. cuTensorNet instead reports -0.008287 and $+0.016849$, so even the direction of the error is backend- and ordering-dependent. The nominal confirm point remains sign-wrong for both Aer orderings and for spectral cuTensorNet. At bond 128 with cutoff 10^{-12} Aer has the correct sign in both orderings, whereas cuTensorNet does so only in sorted order.

The sharpest cross-backend result occurs at bond 1024 and cutoff 10^{-4}: cuTensorNet has the correct sign in both orderings, minimum state fidelity 0.997210, and maximum BKS error 0.000150, while Aer has the wrong sign in both. Only cutoff 10^{-5} gives the correct sign for every tested backend and ordering. At that point cuTensorNet's minimum fidelity is 0.999982 and maximum BKS error is 0.000229; Aer's corresponding minimum fidelity is 0.990609. A named bond/cutoff pair therefore does not define a portable accuracy level across MPS implementations.

Across 20 backend/setting/ordering cohorts, 11 realized signs are correct but only five satisfy the stricter TVD margin certificate; all five are correct. The complete certificate audit is given in the Supplement.

4.7 55-qubit independent backend and classical controls

At spectral ordering and bond 128, independent cuTensorNet MPS yields 15 versus 6 BKS hits for nonlinear versus LR at cutoff 10^{-3} in 5,000 shots per method: rates 0.0030 and 0.0012 (two-sided Fisher exact $p = 0.078$). Tightening to 10^{-4} removes that directional advantage: counts are 11 versus 12, rates 0.0022 and 0.0024 ($p = 1.0$). Near-BKS

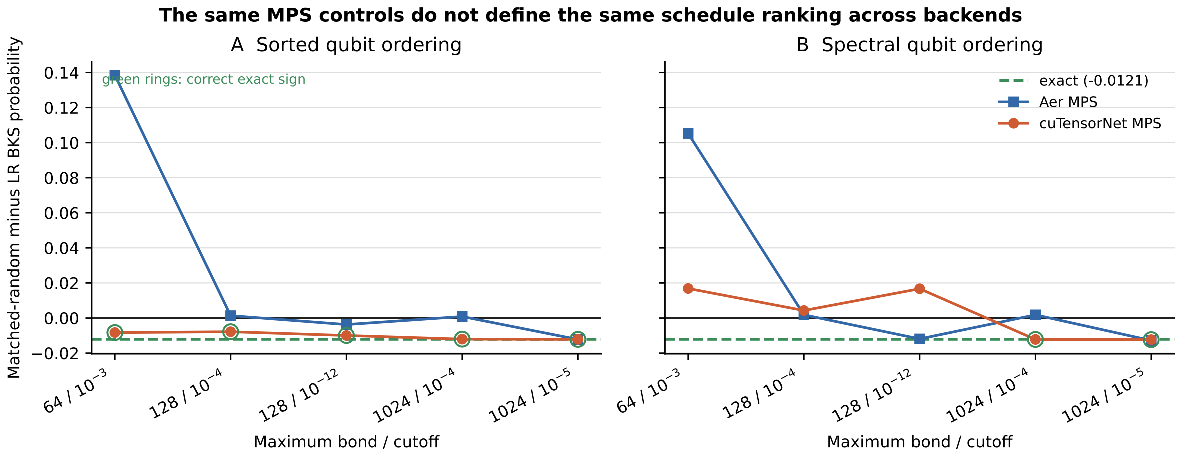

Figure 5: Exact-calibrated matched-random-minus-LR BKS effects on the 24-qubit `aves-sparrow-social` kernel. The dashed line is the dense exact effect; green rings mark cuTensorNet points with its correct sign. Aer and cuTensorNet can disagree in sign at the same nominal bond/cutoff, and qubit ordering changes the approximate ranking despite exact equivalence.

changes from 0.0820 versus 0.0424 to 0.0782 versus 0.0764, while feasible probability remains higher for nonlinear, 0.6232 versus 0.4840 and 0.6352 versus 0.4838. Thus the independent backend does not reproduce a significant LR advantage at the tighter point, but it does reproduce loss of the nonlinear BKS advantage while the feasible ranking remains stable.

The classical comparison rules out a solver-advantage interpretation. HiGHS proves BKS 88 with zero gap in 0.036 seconds. On the full 186-vertex graph, randomized minimum-degree greedy produces 1,133 BKS solutions in 15,000 starts (0.07553) in 19.95 seconds. On the same QOBLIB kernel used by QAOA, it produces 6,063 (0.4042) in 2.21 seconds. These BKS rates are 11.2 and 60.0 times the best QAOA rate. One third of the measured three-method QAOA batch wall time is about 183 seconds, so the corresponding classical heuristic runs are also about 9.2 and 82.9 times faster. The contribution is therefore diagnostic methodology, not quantum or heuristic optimization advantage.

5 Discussion

5.1 Why a rank reversal is plausible

MPS compression repeatedly removes small Schmidt components. BKS probability is a rare-event functional of the final distribution rather than a smooth global norm. A schedule can therefore appear superior because the simulator retains or discards a small set of amplitudes differently. Tightening the threshold changes BKS probability by factors of several for both schedules, yet feasibility changes far less for LR and remains much higher for the nonlinear ramp. This pattern is consistent with rare optimum strings being more sensitive than broad feasible regions. The exact-state calibration supports this mechanism directly: global distribution error falls smoothly as the threshold tightens, while signed BKS error is schedule-specific and crosses zero for LR.

The 55-qubit experiment does not identify which schedule is closer to its exact state. Convergence there would require tighter settings, an independent exact sampler, or quantum-hardware evidence. The five-case exact replication resolves the mechanism across a broader range: large-margin cases remain sign-stable, while failures concentrate where the accumulated event-error budget exceeds the exact schedule margin. The correct general conclusion remains that a BKS ranking is not established by a single approximate setting.

The independent backend strengthens this statement. On the exact-calibrated 24-qubit case, changing implementation or qubit ordering changes the sign at fixed nominal MPS controls. Across all cases, cross-backend sign agreement is 45/50 rather than automatic. The cutoff needed for agreement is therefore an empirical property of the backend–circuit–observable triple, not a portable fidelity specification. Qubit ordering and backend version must be reported alongside the usual MPS truncation controls.

5.2 Implications for optimization benchmarks

We recommend that approximate-simulator submissions report:

1. native circuit, qubit ordering, bond cap, discarded-weight threshold, and whether a basis transpilation was applied;
2. at least three accuracy points bracketing the reported setting;
3. ranking stability for every headline observable, not only average energy or a simulator fidelity proxy;
4. an exact-state or independent-backend anchor whenever

the target size permits one;
5. an observable-level error budget, such as the TVD rank certificate, when a winner is declared by a probability difference;
6. raw counts, shot budgets, and confidence intervals for rare-event metrics; and
7. a distinction between conclusions stable across settings and those conditional on one released benchmark configuration.

These requirements complement QOBLIB's standardized reporting objective [8]. They also matter for transfer learning: an approximate simulator can influence not only final evaluation but validation-time model selection. Our champions were deliberately frozen after a documented code audit, and the first transpiled run remains archived rather than silently deleted.

5.3 Limitations

The transfer study uses one four-instance graph family and one MIS reduction policy. The high-degree pruning step is heuristic, although all reported samples are valid on the original graph and BKS values refer to the full instances. Only two principal schedules receive the full 55-qubit threshold sweep. For that circuit we do not compute state fidelity, discarded-weight accumulation, or exact probabilities. Direct cuTensorNet contraction matches the small exact controls but its 55-qubit exact sampler returns an internal preparation error; the independent large-circuit results remain approximate. The dense replication adds five exact instances but covers five selected MPS points, three frozen schedules, and two software backends on one GPU. Its 100 cohorts are not independent statistical samples of a population: settings and orderings share circuits, so the Fisher test is descriptive and the TVD theorem is the primary support. Both simulators are noiseless and neither constitutes hardware evidence. Finally, the 10,000-shot sensitivity points are single simulator jobs, so their binomial precision is high but job-to-job implementation effects are not separately estimated.

6 Reproducibility

The experiment records Python, Qiskit, Aer, cuTensorNet, NumPy, operating-system metadata, the cloned baseline commit, and a SHA-256 digest of the frozen protocol. Training, validation, blind testing, the discarded transpilation audit, every fidelity point, independent-backend job, and classical run are separate JSON artifacts. Checkpoints are written after long jobs. The test suite verifies the published ramp formula, split disjointness, the 186-to-55 reduction, disabled repair, balanced blind artifacts, exact calibration, classical feasibility, cuTensorNet axis conventions and unit-fidelity controls, the two 5,000-shot points, completeness of the 300-row replication, and NumPy-safe artifact serialization. Analysis scripts regenerate CSV tables and all five figures; export manifests hash every new QPY circuit, exact state, reused artifact, runner, and frozen protocol.

7 Conclusion

On a real 186-vertex QOBLIB instance executed as a 55-qubit, depth-15 QAOA circuit, a transferred nonlinear ramp appears to improve optimum-hit probability by 2.46 times at the released MPS setting, but the ranking reverses when the truncation threshold is tightened. A frozen five-case, 300-row dense replication then shows 91/100 correct signs and only 45/50 cross-backend sign agreements. The observable-level TVD certificate is decisive: all 77 certified cohorts preserve the exact ranking, while every failure lies outside the certificate. Strong classical controls dominate in both success rate and runtime. The broader lesson is methodological, not a claim of quantum advantage. When approximate simulation stands between a variational algorithm and a benchmark claim, convergence of the *ranking observable* is part of the algorithm evaluation, not an optional simulator appendix.

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